

Lab 6 - Recursive Descent Parser

Solved.c

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

int curr = 0;
char str[100];

// -----
void E();
void Eprime();
void T();
void Tprime();
void F();

// -----
void invalid() {
    printf("-----ERROR!-----\n");
    exit(0);
}

void valid() {
    printf("-----SUCCESS!-----\n");
    exit(0);
}

// -----
void E() {
    T();
    Eprime();
}

// -----
void Eprime() {
    if (str[curr] == '+') {
        curr++;
    }
}
```

```
        T();
        Eprime();
    }
}

// -----
void T() {
    F();
    Tprime();
}

// -----
void Tprime() {
    if (str[curr] == '*') {
        curr++;
        F();
        Tprime();
    }
}

// -----
void F() {
    if (str[curr] == '(') {
        curr++;
        E();
        if (str[curr] == ')') {
            curr++;
            return;
        } else {
            invalid();
        }
    } else if (str[curr] == 'i') {
        curr++;
        return;
    } else {
        invalid();
    }
}

// -----
```

```

int main() {
    printf("Enter String: ");
    scanf("%s", str);
    E();
    if (str[curr] == '$') {
        valid();
    } else {
        // printf("%c\n", str[curr]);
        invalid();
    }
}

```

Output:

```

6CSEC1@db1-23:~/Documents/CD_230905152/L6$ cc -o abc ./solved.c
6CSEC1@db1-23:~/Documents/CD_230905152/L6$ ./abc
Enter String: i+i$
-----SUCCESS!-----
6CSEC1@db1-23:~/Documents/CD_230905152/L6$ ./abc
Enter String: hello
-----ERROR!-----

```

Q1.c

```

#include <stdio.h>
#include <stdlib.h>
#include <string.h>

void S();
void T();
void Tp();
void invalid();
void valid();

int curr = 0;
char str[100];

void main() {
    printf("Enter String: ");
    scanf("%s", str);
    S();
    if (str[curr] == '$')
        valid();
    else {
        printf("%c\n", str[curr]);
    }
}

```

```

        invalid();
    }
}

void S() {
    if (str[curr] == 'a' || str[curr] == '>') {
        curr++;
        return;
    }
    if (str[curr] == '(') {
        curr++;
        T();
        if (str[curr] == ')') {
            curr++;
            return;
        }
        invalid();
    }
    invalid();
}

void T() {
    if (str[curr] == 'a' || str[curr] == '>' || str[curr] == '(') {
        S();
        Tp();
        return;
    }
    invalid();
}

void Tp() {
    if (str[curr] == ',') {
        curr++;
        S();
        Tp();
        return;
    }
    if (str[curr] == ')')
        return;
    invalid();
}

void invalid() {
    printf("-----ERROR!-----\n");
}

```

```

        exit(0);
    }

void valid() {
    printf("-----SUCCESS!-----\n");
    exit(0);
}

```

Output:

```

6CSEC1@db1-23:~/Documents/CD_230905152/L6$ cc -o abc ./q1.c
6CSEC1@db1-23:~/Documents/CD_230905152/L6$ ./abc
Enter String: a$
-----SUCCESS!-----
6CSEC1@db1-23:~/Documents/CD_230905152/L6$ ./abc
Enter String: abc$
b
-----ERROR!-----
6CSEC1@db1-23:~/Documents/CD_230905152/L6$ █

```

Q2.c

```

#include <stdio.h>
#include <stdlib.h>
#include <string.h>

void S();
void U();
void V();
void W();

void invalid();
void valid();

int curr = 0;
char str[100];

void main() {
    printf("Enter String: ");
    scanf("%s", str);
    S();
    if (str[curr] == '$')
        valid();
    else {

```

```

        printf("%c\n", str[curr]);
        invalid();
    }
}

void S() {
    if (str[curr] == 'a' || str[curr] == 'd') {
        U();
        V();
        W();
        return;
    }
    invalid();
}

void U() {
    if (str[curr] == 'a') {
        curr++;
        S();
        if (str[curr] == 'b')
            curr++;
        return;
    }
    if (str[curr] == 'd') {
        curr++;
        return;
    }
    if (str[curr] == '(') {
        curr++;
        S();
        if (str[curr] == ')')
            curr++;
        return;
    }
    invalid();
}

void V() {
    if (str[curr] == 'a') {
        curr++;
        V();
        return;
    } else if (str[curr] == 'b' || str[curr] == 'c' || str[curr] == ')') ||
str[curr] == '$')

```

```

        return;
    invalid();
}

void W() {
    if (str[curr] == 'c') {
        curr++;
        W();
        return;
    } else if (str[curr] == 'b' || str[curr] == ')') || str[curr] == '$')
        return;
    invalid();
}

void invalid() {
    printf("-----ERROR!-----\n");
    exit(0);
}

void valid() {
    printf("-----SUCCESS!-----\n");
    exit(0);
}

```

Output:

```

6CSEC1@db1-23:~/Documents/CD_230905152/L6$ cc -o abc ./q2.c
6CSEC1@db1-23:~/Documents/CD_230905152/L6$ ./abc
Enter String: daacc$
-----SUCCESS!-----
6CSEC1@db1-23:~/Documents/CD_230905152/L6$ ./abc
Enter String: why
-----ERROR!-----

```

Q3.c

```

#include <stdio.h>
#include <stdlib.h>
#include <string.h>

void S();
void A();
void Ap();
void B();

```

```

void invalid();
void valid();

int curr = 0;
char str[100];

void main() {
    printf("Enter String: ");
    scanf("%s", str);
    S();
    if (str[curr] == '$')
        valid();
    else {
        printf("%c\n", str[curr]);
        invalid();
    }
}

void S() {
    if (str[curr] == 'a') {
        curr++;
        A();
        if (str[curr] == 'c')
            curr++;
        B();
        if (str[curr] == 'e')
            curr++;
        return;
    }
    invalid();
}

void A() {
    if (str[curr] == 'b') {
        curr++;
        Ap();
        return;
    }
    invalid();
}

void Ap() {
    if (str[curr] == 'b') {

```



```

        curr++;
        Ap();
        return;
    } else if (str[curr] == 'c')
        return;
    invalid();
}

void B() {
    if (str[curr] == 'd') {
        curr++;
        return;
    }
    invalid();
}

void invalid() {
    printf("-----ERROR!-----\n");
    exit(0);
}

void valid() {
    printf("-----SUCCESS!-----\n");
    exit(0);
}

```

Output:

```

6CSEC1@db1-23:~/Documents/CD_230905152/L6$ cc -o abc ./q3.c
6CSEC1@db1-23:~/Documents/CD_230905152/L6$ ./abc
Enter String: abcde$
-----SUCCESS!-----
6CSEC1@db1-23:~/Documents/CD_230905152/L6$ ./abc
Enter String: acde$
-----ERROR!-----

```

Q4.c

```

#include <stdio.h>
#include <stdlib.h>
#include <string.h>

void S(void);
void L(void);
void Lp(void);

```

```

void invalid(void);
void valid(void);

int curr = 0;
char str[100];

int main(void) {
    printf("Enter String: ");
    if (scanf("%99s", str) != 1) {
        invalid();
    }

    S();

    if (str[curr] == '$') {
        valid();
    } else {
        invalid();
    }

    return 0;
}

void S(void) {
    if (str[curr] == '(') {
        curr++;
        L();
        if (str[curr] == ')') {
            curr++;
            return;
        }
        invalid();
    }

    if (str[curr] == 'a') {
        curr++;
        return;
    }

    invalid();
}

void L(void) {

```

```

    if (str[curr] == '(' || str[curr] == 'a') {
        S();
        Lp();
        return;
    }

    invalid();
}

void Lp(void) {
    if (str[curr] == ',') {
        curr++;
        S();
        Lp();
        return;
    }

    if (str[curr] == ')') {
        return;
    }

    invalid();
}

void invalid(void) {
    printf("-----ERROR!-----\n");
    exit(0);
}

void valid(void) {
    printf("-----SUCCESS!-----\n");
    exit(0);
}

```

Output:

```

6CSEC1@db1-23:~/Documents/CD_230905152/L6$ cc -o abc ./q4.c
6CSEC1@db1-23:~/Documents/CD_230905152/L6$ ./abc
Enter String: (a,a)$
-----SUCCESS!-----
6CSEC1@db1-23:~/Documents/CD_230905152/L6$ ./abc
Enter String: (a,$
-----ERROR!-----

```

